

Final Mock Exam — Solutions

Lectures 1–12 (comprehensive)

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Overview of the final mock exam

Problem	Topic	Points
P1	Moving beyond linearity — splines (Ch. 7)	16
P2	Generalised additive models (Ch. 7)	8
P3	Regression and classification trees by hand (Ch. 8)	20
P4	Ensembles — bagging, random forests, boosting (Ch. 8)	16
P5	Neural networks by hand (Ch. 10)	20
P6	Multiple testing (Ch. 13)	20
P7	Cumulative essentials (Ch. 2–6)	20
Σ		120

Exam conditions

120 minutes for 120 points (≈ 1 point per minute). Allowed aids: non-programmable calculator and one handwritten A4 sheet (both sides). All required constants (e-powers and logs) are given in the problems. Justify every answer.

Problem 1 — task

Problem 1 (16 P) — Moving beyond linearity

- (a) [4 P] Degrees of freedom of a cubic spline with K interior knots — justify by counting parameters and constraints; evaluate for $K = 4$.
- (b) [4 P] Degrees of freedom of a *natural* cubic spline with K knots ($K = 4$); which extra constraint acts where and why is it useful?
- (c) [4 P] Smoothing spline $\sum_i (y_i - g(x_i))^2 + \lambda \int g''(t)^2 dt$: role of the two terms; behaviour of $\hat{g}$ as $\lambda \rightarrow 0$ and $\lambda \rightarrow \infty$.
- (d) [4 P] Truncated-power basis of a cubic spline with one knot ξ ; define $(x - \xi)_+^3$; why does adding it keep the fit smooth at ξ ?

Problem 1 — solution (a) and (b)

Solution P1 — (a) and (b)

(a) K knots $\Rightarrow K + 1$ cubic pieces $\times 4$ coefficients $= 4K + 4$ parameters; continuity of g and g' and g'' at each knot $\Rightarrow 3K$ constraints:

$$\text{df} = 4(K + 1) - 3K = K + 4 \quad \Rightarrow \quad K = 4 : \text{df} = 8.$$

(Equivalently: basis $1, x, x^2, x^3$ plus one truncated power per knot.)

(b) Natural spline: *linear beyond the boundary knots* (regions $X < \xi_1$ and $X > \xi_K$); killing the quadratic and cubic terms there costs 2 parameters per boundary region:

$$\text{df} = (K + 4) - 4 = K \quad \Rightarrow \quad K = 4 : \text{df} = 4.$$

Why useful: splines are most erratic near the boundaries where data are sparse; the linearity constraint stabilises the fit there (lower variance for a little extra bias).

Problem 1 — solution (c) and (d)

Solution P1 — (c) and (d)

(c) RSS term = fidelity to the data; $\lambda \int g''(t)^2 dt$ penalises curvature (wiggleness); λ trades the two off.

- $\lambda \rightarrow 0$: no penalty $\Rightarrow \hat{g}$ *interpolates* the training points exactly — extremely wiggly, zero training RSS, severe overfitting (effective df $\rightarrow n$).
- $\lambda \rightarrow \infty$: curvature infinitely expensive $\Rightarrow g'' \equiv 0 \Rightarrow \hat{g}$ is the *least-squares straight line* (effective df $\rightarrow 2$).

λ (chosen by LOOCV) steers the effective df smoothly between n and 2.

(d) $f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 (x - \xi)_+^3$ with

$$(x - \xi)_+^3 = \begin{cases} (x - \xi)^3 & x > \xi \\ 0 & x \leq \xi \end{cases}$$

$(x - \xi)_+^3$ has value and first and second derivative 0 at $\xi \Rightarrow$ only the third derivative jumps: the fit stays continuous with continuous g' and g'' — a cubic spline.

Problem 2 — task

Problem 2 (8 P) — Generalised additive models

- (a) [3 P] General form of a GAM for quantitative Y ; how does it extend multiple linear regression?
- (b) [3 P] One advantage concerning the interpretation of the f_j ; the main structural limitation — and its repair.
- (c) [2 P] Backfitting in one or two sentences.

Problem 2 — solution

Solution P2 — GAMs

(a) $Y = \beta_0 + f_1(X_1) + f_2(X_2) + \dots + f_p(X_p) + \varepsilon$.

Linear regression is the special case $f_j(X_j) = \beta_j X_j$; a GAM replaces each linear term by a smooth non-linear f_j (spline, local regression) but keeps the *additive* structure.

(b) *Advantage*: additivity $\Rightarrow$ the effect of each X_j is its own function f_j , which can be plotted and interpreted individually while the other predictors are held fixed — one panel per predictor, as readable as a coefficient table.

Limitation: no *interactions* — the effect of X_j cannot depend on X_k .

Repair: add interaction components manually, e.g. $X_j \times X_k$ or a two-dimensional smooth $f_{jk}(X_j, X_k)$.

(c) *Backfitting*: cycle through $j = 1, \dots, p$ and re-fit each f_j to the partial residuals $r_j = y_i - \beta_0 - \sum_{k \neq j} f_k(x_{ik})$ with all other functions held fixed; repeat until convergence.

Problem 3 — task

Problem 3 (20 P) — Trees by hand

Advertising budget x (1,000 EUR) and weekly sales y (1,000 units) for $n = 6$ products; one split at s : $R_1 = \{x < s\}$ and $R_2 = \{x \geq s\}$ with region-mean predictions.

i	1	2	3	4	5	6
x_i	1	2	3	4	5	6
y_i	2	4	3	3	8	10

(a) **[8 P]** Evaluate the cutpoints $s = 2.5$ and $s = 4.5$ by total RSS; which is chosen? (b) **[2 P]** Prediction for a new product with $x = 5$. (c) **[6 P]** Node with 6 “Yes” / 2 “No”: proportions; Gini; entropy ($\ln 0.75 = -0.2877$; $\ln 0.25 = -1.3863$); predicted class. (d) **[4 P]** Cost-complexity pruning: role of α in $\sum_m \sum_{i \in R_m} (y_i - \hat{y}_{R_m})^2 + \alpha |T|$.

Problem 3 — solution (a) and (b)

Solution P3 — (a) and (b)

(a) Region means and RSS per cutpoint:

s	R_1 (y-values)	$\hat{y}_{R_1}$	R_2 (y-values)	$\hat{y}_{R_2}$	total RSS
2.5	{2,4}	3	{3,3,8,10}	6	$2 + 38 = 40$
4.5	{2,4,3,3}	3	{8,10}	9	$2 + 2 = 4$

Details: $s = 2.5$: $(2 - 3)^2 + (4 - 3)^2 = 2$; $(3 - 6)^2 + (3 - 6)^2 + (8 - 6)^2 + (10 - 6)^2 = 9 + 9 + 4 + 16 = 38$.

$s = 4.5$: $1 + 1 + 0 + 0 = 2$; $(8 - 9)^2 + (10 - 9)^2 = 2$.

Greedy recursive binary splitting picks the smaller RSS: $s = 4.5$ ($4 \ll 40$; root RSS was 52) — it isolates the two high-sales products.

(b) $x = 5 \geq 4.5 \Rightarrow$ region $R_2 \Rightarrow \hat{y} = \hat{y}_{R_2} = 9$ (i.e. 9,000 units per week).

Problem 3 — solution (c) and (d)

Solution P3 — (c) and (d)

(c) $\hat{p}_{\text{Yes}} = 6/8 = 0.75$; $\hat{p}_{\text{No}} = 2/8 = 0.25$.

$$G = 0.75 \cdot 0.25 + 0.25 \cdot 0.75 = 0.375$$

$$D = -(0.75 \ln 0.75 + 0.25 \ln 0.25) = 0.75 \cdot 0.2877 + 0.25 \cdot 1.3863 = 0.2158 + 0.3466 = 0.562$$

Predicted class: majority = **Yes** (estimated probability 0.75). Both measures are near 0 for pure nodes — this node is moderately impure.

(d) α = price per terminal node $|T|$:

- $\alpha = 0$: criterion = training RSS $\Rightarrow$ full tree T_0 is optimal.
- $\alpha \uparrow$: leaves must pay for themselves; branches with small RSS gain are pruned $\Rightarrow$ nested sequence of ever smaller subtrees (weakest-link pruning).
- Choice: *cross-validation* over α ; refit the chosen subtree on all data — balances bias against variance.

Problem 4 — task

Problem 4 (16 P) — Ensembles

- (a) [6 P] B identically distributed trees with variance σ^2 and pairwise correlation ρ :
 $\text{Var}(\frac{1}{B} \sum_b \hat{f}_b(x)) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$. (i) Why does this show variance reduction and what limits it? (ii) Evaluate for $B = 100$ and $\rho = 0.6$ and $\sigma^2 = 4$; compare with one tree. (iii) Limit as $B \rightarrow \infty$?
- (b) [4 P] Random forests: modification at each split; recommended m for $p = 25$; why does it improve on bagging (use the formula)?
- (c) [4 P] Boosting: the three tuning parameters; the slow-learning idea.
- (d) [2 P] What does the OOB error estimate and why is it free?

Problem 4 — solution (a) and (b)

Solution P4 — (a) and (b)

(a) (i) Averaging drives the second term $\frac{1-\rho}{B}\sigma^2 \rightarrow 0$ (the independent part of the trees' variability); the first term $\rho\sigma^2$ is independent of B — correlation between bootstrap trees limits the reduction.

(ii) $\text{Var} = 0.6 \cdot 4 + \frac{0.4}{100} \cdot 4 = 2.4 + 0.016 = 2.416$ vs. $\sigma^2 = 4$ for a single tree ($\approx 40\%$ less).

(iii) $B \rightarrow \infty$: variance $\rightarrow \rho\sigma^2 = 2.4$ (floor; more trees never overfit, they only stabilise).

(b) At *each split* only a random subset of m of the p predictors may be used; classification default $m \approx \sqrt{p} \Rightarrow p = 25$: $m = \sqrt{25} = 5$.

With bagging a dominant predictor tops nearly every tree $\Rightarrow$ trees alike $\Rightarrow \rho$ high. Restricting candidates *decorrelates* the trees ($\rho \downarrow$), which lowers the floor $\rho\sigma^2$ — the part of the variance that averaging alone cannot remove.

Problem 4 — solution (c) and (d)

Solution P4 — (c) and (d)

(c) Boosting for regression trees:

- B — **number of trees**: added sequentially; too many $\Rightarrow$ slow overfitting; choose by CV.
- λ — **shrinkage** (e.g. 0.01 or 0.001): scales each new tree's contribution; smaller $\lambda \Rightarrow$ slower learning $\Rightarrow$ larger B needed.
- d — **splits per tree** (interaction depth): complexity and interaction order of each tree; often stumps ($d = 1$) suffice (additive model).

Slow learning: each tree is fitted to the current *residuals* and only the fraction λ is added — the model improves gradually exactly where it still errs, instead of fitting the data hard at once; this tends to generalise better.

(d) Each bootstrap sample misses $\approx 1/3$ of the observations (out-of-bag); predicting every observation only with trees that never saw it yields the **OOB error** — a valid *test-error* estimate, free as a by-product of the bootstrap (no validation set or CV loop needed).

Problem 5 — task

Problem 5 (20 P) — Neural networks by hand

Input $x = (1, 2)$; hidden ReLU units $g(z) = \max\{0, z\}$; linear output:

$$A_1 = g(-1 + 1x_1 + 2x_2) \quad A_2 = g(-3 - 2x_1 + 1x_2) \quad f(x) = 1 + 2A_1 + 4A_2$$

- (a) **[8 P]** Forward pass step by step: pre-activations; A_1 and A_2 ; output $f(x)$. What did the ReLU do to unit 2 and why are non-linear activations essential?
- (b) **[6 P]** Logits $z = (1.0, 0.0, -1.0)$; true class 1: all three softmax probabilities and the cross-entropy loss ($e^1 = 2.718$; $e^0 = 1.000$; $e^{-1} = 0.368$; $\ln 4.086 = 1.408$).
- (c) **[4 P]** Parameter count of a $p = 4 \rightarrow k = 3 \rightarrow K = 2$ network with biases (show both layers).
- (d) **[2 P]** Conv layer: 32×32 input; $F = 5$; $P = 2$; $S = 1$: output size via $(W - F + 2P)/S + 1$.

Problem 5 — solution (a) forward pass

Solution P5 — (a) forward pass

Pre-activations:

$$z_1 = -1 + 1 \cdot 1 + 2 \cdot 2 = 4$$

$$z_2 = -3 - 2 \cdot 1 + 1 \cdot 2 = -3$$

ReLU activations:

$$A_1 = \max\{0, 4\} = 4$$

$$A_2 = \max\{0, -3\} = 0$$

Output:

$$f(x) = 1 + 2 \cdot 4 + 4 \cdot 0 = 9$$

The ReLU *switched unit 2 off*: its negative pre-activation is clipped to 0, so it contributes nothing for this input (though it can for others).

Why essential: with a linear g the whole network collapses into one linear function of x regardless of depth — the non-linearity is what lets hidden layers build flexible functions (piecewise-linear surfaces with ReLU).

Problem 5 — solution (b) softmax and cross-entropy

Solution P5 — (b) softmax and cross-entropy

Exponentials and their sum:

$$e^{1.0} = 2.718 \quad e^{0.0} = 1.000 \quad e^{-1.0} = 0.368 \quad \Rightarrow \quad 2.718 + 1.000 + 0.368 = 4.086$$

Softmax probabilities:

$$\hat{p}_1 = \frac{2.718}{4.086} = 0.665 \quad \hat{p}_2 = \frac{1.000}{4.086} = 0.245 \quad \hat{p}_3 = \frac{0.368}{4.086} = 0.090$$

(sum = 1.000 ✓). Cross-entropy for true class 1:

$$-\ln \hat{p}_1 = -\ln \frac{e^1}{4.086} = -(1 - \ln 4.086) = 1.408 - 1 = 0.408$$

Sanity checks: a perfect prediction ($\hat{p}_1 = 1$) gives loss 0; predicting the uniform $1/3$ would give $\ln 3 \approx 1.099$ — here the network is right and fairly confident.

Problem 5 — solution (c) and (d)

Solution P5 — (c) parameters and (d) convolution

(c) Fully connected $4 \rightarrow 3 \rightarrow 2$ with biases:

Layer	Weights	Biases	Parameters
input $\rightarrow$ hidden	$4 \cdot 3 = 12$	3	15
hidden $\rightarrow$ output	$3 \cdot 2 = 6$	2	8
total			23

(d) Output width = $\frac{W - F + 2P}{S} + 1 = \frac{32 - 5 + 2 \cdot 2}{1} + 1 = 31 + 1 = 32 \Rightarrow$ output feature map 32×32 :
with $F = 5$ and $P = 2$ and $S = 1$ this is the “same” convolution — spatial size preserved (one map per filter).

Problem 6 — task

Problem 6 (20 P) — Multiple testing

- (a) **[4 P]** $m = 20$ independent tests at $\alpha = 0.05$; all nulls true: define and compute the FWER $((0.95)^{20} = 0.3585)$; interpret.
- (b) **[7 P]** Six ordered p -values 0.001; 0.004; 0.011; 0.030; 0.045; 0.230 — FWER control at $\alpha = 0.05$ with (i) Bonferroni and (ii) Holm; thresholds; rejected hypotheses; why is Holm never weaker than Bonferroni?
- (c) **[6 P]** Benjamini–Hochberg at $q = 0.05$ on the same p -values: thresholds jq/m ; rejections; what exactly does FDR control guarantee?
- (d) **[3 P]** Exploratory screen of 20,000 genes with follow-up validation: FWER or FDR? Justify.

Problem 6 — solution (a) FWER

Solution P6 — (a) FWER for 20 independent tests

$\text{FWER} = \Pr(V \geq 1)$ — the probability of *at least one* false rejection among the m tests (V = number of falsely rejected true nulls).

With all 20 nulls true and independent tests at level 0.05:

$$\text{FWER} = 1 - \Pr(\text{no false rejection}) = 1 - (1 - 0.05)^{20} = 1 - (0.95)^{20} = 1 - 0.3585 = 0.6415$$

Interpretation: although each test individually errs with probability 5% only, the chance of at least one spurious “discovery” among 20 unadjusted tests is about 64% — multiple testing without adjustment almost guarantees false findings (with $m = 100$: $1 - 0.95^{100} \approx 0.994$).

Problem 6 — solution (b) Bonferroni and Holm

Solution P6 — (b) Bonferroni and Holm at level 0.05

Bonferroni: reject iff $p_{(j)} \leq \alpha/m = 0.05/6 = 0.00833$: only 0.001 and 0.004 pass ($0.011 > 0.00833$) $\Rightarrow$ reject $H_{(1)}$ and $H_{(2)}$: **2 rejections**.

Holm (step-down): compare $p_{(j)}$ with $\alpha/(m - j + 1)$; stop at first failure:

j	1	2	3	4	5	6
$p_{(j)}$	0.001	0.004	0.011	0.030	0.045	0.230
$\alpha/(m - j + 1)$	0.00833	0.0100	0.0125	0.0167	0.0250	0.0500
decision	✓	✓	✓	stop	—	—

$0.011 \leq 0.0125$ but $0.030 > 0.0167 \Rightarrow$ reject $H_{(1)}$ and $H_{(2)}$ and $H_{(3)}$: **3 rejections**.

Holm $\supseteq$ Bonferroni: Holm's hurdles $\alpha/(m - j + 1) \geq \alpha/m$ for all j (equality only at $j = 1$) $\Rightarrow$ uniformly more powerful — and it still controls the FWER without independence assumptions.

Problem 6 — solution (c) and (d)

Solution P6 — (c) Benjamini–Hochberg and (d) FDR vs FWER

(c) Thresholds $jq/m = j \cdot 0.05/6$:

j	1	2	3	4	5	6
$p_{(j)}$	0.001	0.004	0.011	0.030	0.045	0.230
jq/m	0.00833	0.0167	0.0250	0.0333	0.0417	0.0500
$p_{(j)} \leq jq/m?$	yes	yes	yes	yes	no	no

Largest such j : $L = 4$ ($0.030 \leq 0.0333$; note $0.045 > 0.0417$) $\Rightarrow$ reject *all* of $H_{(1)}$ to $H_{(4)}$: **4 rejections**.
Guarantee: $\text{FDR} = E[V/\max(R,1)] \leq q$ — *on average* at most 5% of rejected hypotheses are false discoveries (no bound on $\Pr(V \geq 1)$; that is FWER).

(d) Control the **FDR**. Bonferroni at $0.05/20,000 = 2.5 \cdot 10^{-6}$ would leave almost no power — guarding against a *single* false positive is needlessly strict when every hit is re-tested anyway. FDR control tolerates a small known fraction of false leads in exchange for a much richer list of candidates — exactly right for exploratory screening with follow-up validation.

Problem 7 — task

Problem 7 (20 P) — Cumulative essentials

- (a) [4 P] True or false: “more flexible always means lower *test* error”. Justify.
- (b) [4 P] Logistic regression for default: $\hat{\beta}_1 = 0.7$ for missed payments ($e^{0.7} = 2.014$): effect of one more missed payment on the *odds*; why is there no single number for the effect on the *probability*?
- (c) [4 P] One reason to prefer 5- or 10-fold CV over LOOCV and one over a single validation split.
- (d) [2 P] $p = 2,000$ predictors; only a handful truly matter: ridge or lasso? Why?
- (e) [6 P] Match to {linear regression; lasso; logistic regression; random forest} — each exactly once: (i) price per m² with CI; (ii) drug response from 5,000 genes ($n = 100$) with a short gene list; (iii) default yes/no with odds reported to the regulator; (iv) machine failures from hundreds of interacting sensors; accuracy only.

Problem 7 — solution (a) to (c)

Solution P7 — (a) to (c)

(a) False. Test error = $\text{bias}^2 + \text{variance} + \text{irreducible error}$: flexibility lowers bias but raises variance; past the sweet spot the test error *rises* (U-shape). E.g. near-linear truth or small n : linear regression beats a flexible overfitter. (Only *training* error always falls with flexibility.)

(b) One more missed payment adds 0.7 to the log-odds, i.e. *multiplies* the odds of default by $e^{0.7} = 2.014$ (roughly doubles them) — the same factor at every x . The *probability* change depends on the position on the S-curve: large near $p \approx 0.5$, tiny near 0 or 1 $\Rightarrow$ no single number.

(c) vs. *LOOCV*: the n leave-one-out fits are nearly identical $\Rightarrow$ their errors are highly correlated $\Rightarrow$ the LOOCV average has *higher variance* (and costs n fits). vs. *validation split*: a single split is highly variable and overestimates test error since only part of the data trains the model (*bias*); k -fold trains on $(k-1)/k$ of the data and averages k estimates $\Rightarrow$ less bias and more stability.

Problem 7 — solution (d) and (e)

Solution P7 — (d) and (e)

(d) Lasso: the ℓ_1 penalty sets coefficients *exactly to zero* $\Rightarrow$ variable selection; ideal for a sparse truth (a handful of signals among 2,000) and yields a small interpretable model. Ridge keeps all 2,000 predictors.

(e) Matching (each method once):

- (i) **Linear regression** — quantitative response; goal is *inference*: effect size with confidence interval from OLS standard errors.
- (ii) **Lasso** — $p = 5,000 \gg n = 100$: OLS infeasible; ℓ_1 selects a short list of genes.
- (iii) **Logistic regression** — binary response; coefficients give odds ratios $e^{\hat{\beta}_j}$: interpretable for the regulator.
- (iv) **Random forest** — automatic non-linearities and interactions; strong accuracy with many mixed predictors; interpretability not needed.